Systems & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

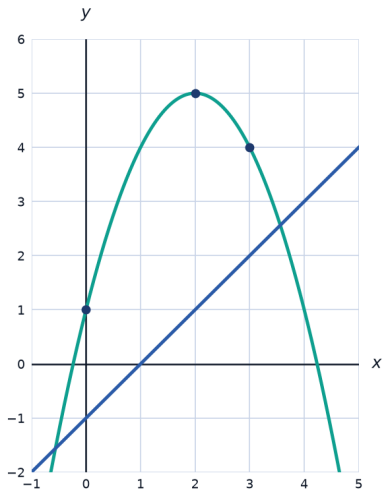
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Geometric sequences as exponential functions**

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

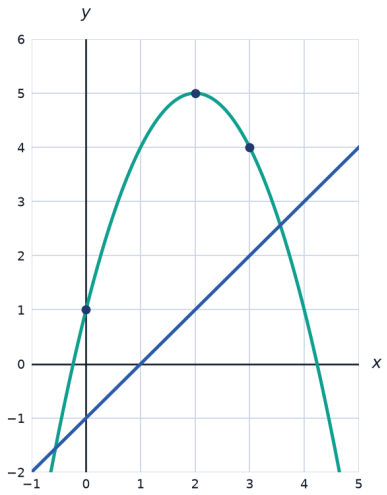
2 **[Foundational]** • **Target: Geometric sequences as exponential functions**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



3 **[Foundational]** • **Target: Geometric sequences as exponential functions**

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



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4 [Foundational] • Target: Geometric sequences as exponential functions

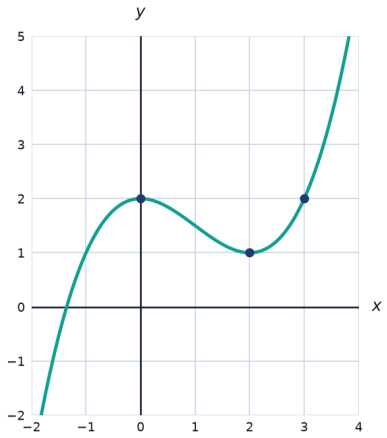
A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

5 [Foundational] • Target: Geometric sequences as exponential functions

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

6 [Foundational] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(2)$.



7 [Foundational] • Target: Comparing exponential relationships

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing exponential relationships

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

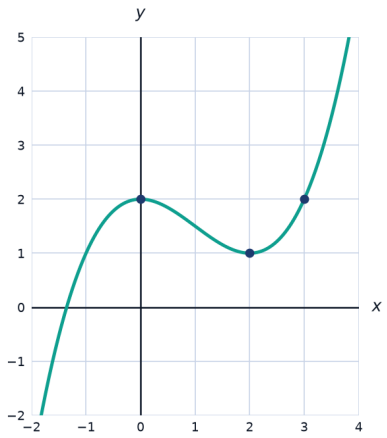
9 [Intermediate] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(3)$.

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10 **[Intermediate]** • **Target: Comparing exponential relationships**

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



11 **[Intermediate]** • **Target: Comparing exponential relationships**

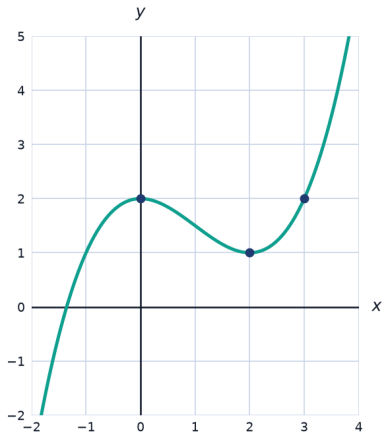
For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(4)$.

12 **[Intermediate]** • **Target: Comparing exponential relationships**

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

13 **[Intermediate]** • **Target: Comparing exponential relationships**

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(5)$.



14 **[Intermediate]** • **Target: Comparing exponential relationships**

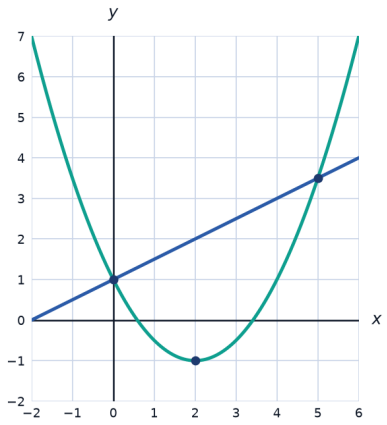
Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(2)$.



16 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 11.01 Statistics & Data Analysis

A data set recorded for comparing linear and exponential relationships consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

17 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 2.01 Quadratic Functions

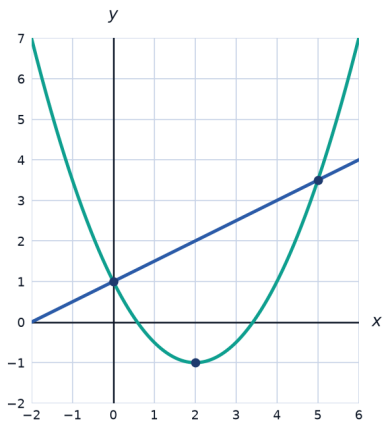
For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(3)$.

18 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(4)$.

19 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 4.01 Functions & Transformations

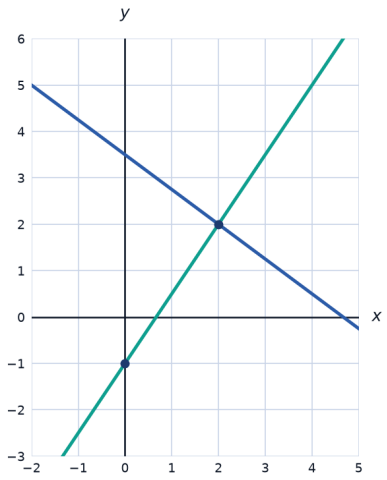
For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(5)$.



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20 [Challenging] • Target: Simple interest • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

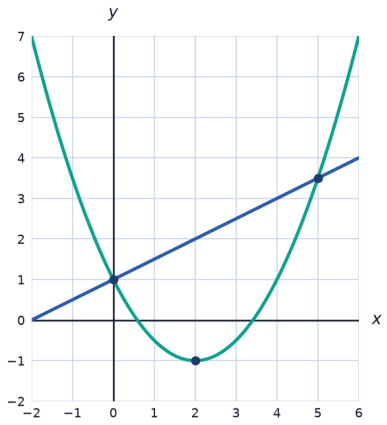
For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(2)$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.



a. 19

b. 24

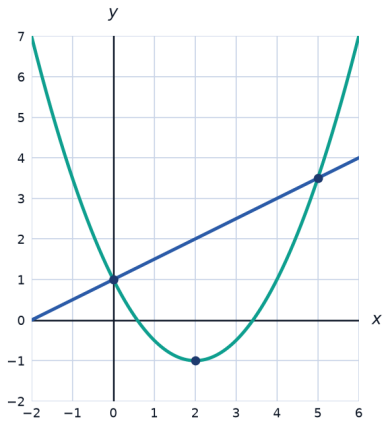
c. 10

d. 23

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22 [Foundational] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

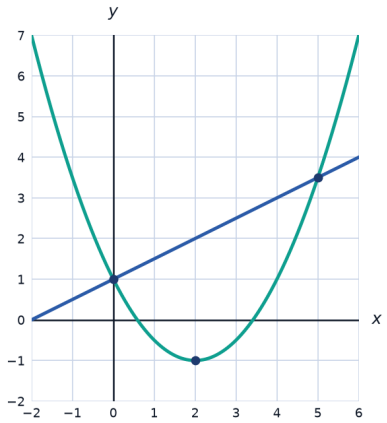
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

23 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(5)$.



a. 34

b. 39

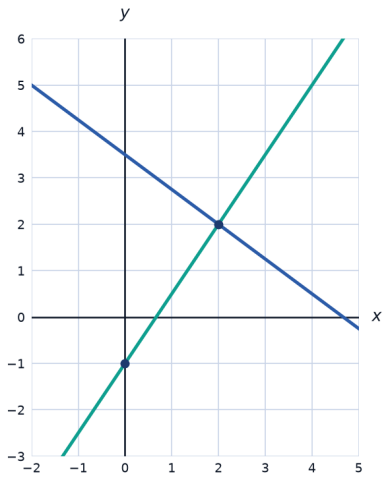
c. 25

d. 38

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24 [Foundational] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.



a. 15

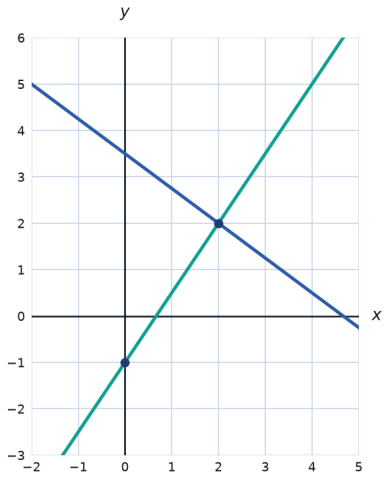
b. 21

c. 12

d. 19

25 [Foundational] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

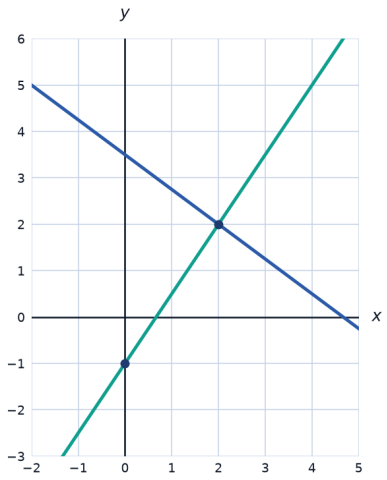
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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26 [Foundational] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(5)$.



a. 33

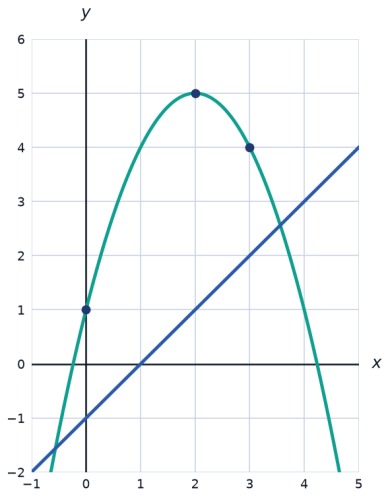
b. 39

c. 30

d. 37

27 [Foundational] • Target: Geometric sequences as exponential functions

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 105°

b. 115°

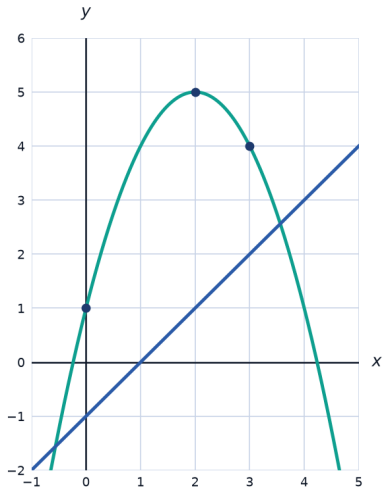
c. 97°

d. 150°

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28 [Foundational] • Target: Geometric sequences as exponential functions

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?



a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Graphs and characteristics of exponential functions

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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31 [Intermediate] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(3)$.

a. 24

b. 29

c. 15

d. 28

32 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

33 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

34 [Intermediate] • Target: Exponential growth and decay

A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

35 [Intermediate] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(3)$.

a. 21

b. 27

c. 18

d. 25

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36 [Intermediate] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

37 [Intermediate] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 3.01 Exponents & Polynomials

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 10

c. 8

d. 12

39 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 4.01 Functions & Transformations

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98°

b. 108°

c. 90°

d. 150°

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40 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 91°

b. 101°

c. 83°

d. 150°

41 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 7.01 Sequences & Series

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

42 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 8.01 Factoring Polynomials

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20

d. 24

43 [Challenging] • Target: Comparing exponential relationships • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

44 [Challenging] • Target: Simple interest • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for simple interest consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

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45 [Challenging] • Target: Simple interest • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(3)$.

a. 22

b. 27

c. 15

d. 26

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